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Programmable FPGA-based TDLAS-WMS hygrometer (PFTWH) for broad-range water vapour detection: Quantitative assessment of digital lock-in amplifier (DLIA) performance via integrated WMS system (I-WMS)

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ABSTRACT

This study demonstrates the capability of wavelength modulation spectroscopy (WMS) based on tunable diode laser absorption spectroscopy (TDLAS) using a hand-held field-programmable gate array (FPGA) system. Programmable FPGA-based TDLAS-WMS hygrometer (PFTWH) enables measurement of water vapour concentrations over a broad range (70–5800 ppm) for applications in climate science. This system incorporates a custom-designed digital lock-in amplifier (DLIA), optimized for water vapour detection. For the DLIA, a 100 Hz cut-off frequency and a finite impulse response filter with a windowed-Hamming design and a 200-tap coefficient were used. The PFTWH demonstrated simultaneous processing capability of 1 f and 2 f signals by programming the DLIA on a commercial FPGA board. To quantitatively evaluate DLIA performance, we introduced an integrated WMS system that enables parallel operation of a commercial FPGA-based TDLAS-WMS hygrometer (CFTWH), the PFTWH, and direct TDLAS (dTDLAS). The 1 f and 2 f signals are extracted from both the CFTWH and PFTWH, while the dTDLAS provides reference concentration information for performance assessment. The linearity and accuracy of the designed PFTWH were validated by comparison with the CFTWH. Additionally, long-term measurements of water vapour concentrations across the broad range (70–5800 ppm) were successfully performed with a time resolution of 0.1 s, maintaining stable performance over 1000 s with 30 s averaging. Furthermore, both PFTWH and CFTWH were demonstrated to meet uncertainty, accuracy, and precision requirements for airborne measurement.

1. Introduction

Accurately predicting Earth's climate system requires a deep understanding of the integrated water cycle. A key focus in climate research is how atmospheric moisture influences cloud formation and different precipitation phases [1]. Observing changes in water vapour concentration with altitude is essential for studying atmospheric processes [2]. Since the majority of water vapour resides in the troposphere [3], vertical humidity profiles and turbulence measurements are primarily conducted within this layer [4]. Although the stratosphere contains lower concentrations of water vapour, it still plays a crucial role in the radiative and chemical dynamics of the upper troposphere and contributes to stratospheric equilibrium [5]. Therefore, obtaining precise measurements across a broad range of water vapour concentrations is necessary [6].

Both remote sensing and in-situ hygrometer are fundamental for atmospheric studies; however, in-situ methods are indispensable due to the spatial resolution and vertical range limitations of remote sensing techniques. Among various in-situ measurement approaches, tunable diode laser absorption spectroscopy (TDLAS) is particularly well-suited

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for humidity and trace gas monitoring. This technique employs a tunable diode laser to scan across an absorption line with an extremely narrow bandwidth [7]. By varying the laser current, the wavelength is finely adjusted to capture the absorption spectrum of specific gases. The area under the measured absorption spectrum is directly proportional to gas concentration [8], enabling precise quantification through signal processing.

However, in some cases, the absorption spectrum may not be fully captured, or high noise levels may reduce measurement accuracy and reliability [9]. To overcome these challenges, wavelength modulation spectroscopy (WMS) is suggested for accurate and sensitive measurement in harsh environment [10–15].

Previous studies on TDLAS-WMS have implemented WMS using commercial lock-in amplifier (LIA) and data acquisition devices [16–18]. Unlike direct absorption spectroscopy, which records a transmission spectrum, TDLAS-WMS extracts harmonic signals through an LIA [19–24]. However, LIAs are expensive and unsuitable for in-situ applications, while data processing is hindered by large data streams and long processing times. As a result, traditional gas sensing systems are often bulky and power-intensive.

To address these challenges, some industrial TDLAS sensing systems have incorporated self-developed laser drivers and LIAs [25,26], though these remained as separate modules, increasing system size. A more effective alternative is the use of field-programmable gate array (FPGA) for real-time gas and temperature measurements.

FPGAs play a crucial role in advancing modern signal processing algorithms by enhancing frequency resolution, improving phase accuracy, and expanding operational bandwidths. They provide a flexible platform for implementing customized digital LIA architectures, integrating direct digital synthesizer (DDS) for precise frequency generation, adaptive filters for noise reduction, and phase-sensitive detection (PSD) algorithms for high-performance operation [27]. Additionally, FPGAs enable efficient resource utilization with minimal hardware overhead while maintaining real-time processing capabilities. Analog designs often suffer from error sources that are difficult to mitigate due to inherent hardware limitations, whereas digital implementations allow for error correction through modifications in functional blocks or signal enhancement algorithms, thereby improving accuracy. FPGA-based digital LIAs offer significant advantages for weak signal detection, benefiting applications in electronic science, signal processing, and sensor technologies [28].

Since FPGA provides a programmable platform capable of real-time data manipulation, featuring extensive logic resources and block random access memory (BRAM), when properly synthesized, they can perform complex mathematical computations at clock speeds ranging from 40 MHz to 1 GHz, enabling efficient data acquisition and processing of WMS signals. Although WMS inherently suppresses noise, analyzing the harmonic components generated by the interaction between the modulation signal and absorption features remains challenging due to potential interference from laser fluctuations, plumes, or electronic noise [29]. To overcome these challenges, digital lock-in amplifiers (DLIAs) - combining PSD and low-pass filter with a fixed system clock - are widely employed in instrumentation. These DLIAs can accurately detect and isolate low-amplitude signals affected by noise and interference, facilitating precise extraction of 1 f and 2 f harmonic components from light incident on the photodiode. This function is fundamental to signal processing in TDLAS-WMS [30]. It is reported that FPGA-based DLIA system enhances the sensitivity of one order of magnitude more than analog LIA-based work in case of CO sensing [14, 15]

While several studies [31–34] have investigated the theoretical integration of FPGA processors in TDLAS, only a few [5], [35] have demonstrated their practical implementation in humidity or gas monitoring systems. Although these efforts aim to fully miniaturize FPGA-based gas monitoring systems, they are still in the early stages and are not yet fully functional. Given the current development of FPGA-based analysis systems, we believe it is more effective to break down and examine each FPGA component individually to evaluate its contribution. This method allows for a more detailed feasibility assessment compared to existing systems and paves the way toward the development of fully miniaturized, portable solutions.

As DLIA is recognized as one of the most valuable features for gas monitoring within FPGA processors, this work focuses on quantitatively assessing DLIA performance in TDLAS-WMS systems using a hand-held FPGA board in integrated WMS (I-WMS) setup, as illustrated in Fig. 1. Two configurations were investigated: (1) a commercial FPGA-based signal analyzer with DLIA, referred to as the commercial FPGA-TDLAS-WMS hygrometer (CFTWH), and (2) a programmable FPGA-based board system with DLIA, referred to as the programmable FPGA-TDLAS-WMS hygrometer (PFTWH).

To enable a fair and systematic comparison, I-WMS was devised to evaluate the performance of both CFTWH and PFTWH with respect to the reference direct TDLAS (dTDLAS) method. In particular, the focus is placed on the DLIA performance of PFTWH.

The proposed PFTWH processes wavelength-modulated signals emitted by a single near-infrared distributed-feedback (DFB) laser scanning the water vapour absorption feature near 1370 nm. The FPGA-based electronic system digitally demodulates the WMS signals, and the normalized WMS-2 f/1 f is applied to enhance the spectrometer's sensitivity and robustness. Through this platform, the DLIA performance of PFTWH is quantitatively evaluated and correlated with sensing performance.

2. Methodology

2.1. TDLAS and WMS

TDLAS is a method for measuring gas concentration by passing a laser with finely tunable wavelengths through a gas and analyzing the absorption spectrum that appears. The gas concentration can be calculated from the measured absorption spectrum based on the Beer-Lambert law [36].

$$I(\nu) = I_0(\nu)\cdot\exp[-k(\nu)\cdot L] \tag{1}$$

Here, (ν) represents the intensity of light absorbed at frequency ν, $I_0(\nu)$ is the initial intensity of light before absorption, $k(\nu)$ is the absorption coefficient (cm^{-1}), and L is the path length of the absorbing medium (cm).

The optical density, $OD(\nu)$, can be expressed in terms of (ν) and $I_0(\nu)$ as follows:

$$OD(\nu) = -\ln[I(\nu)/I_0(\nu)] = k(\nu)\cdot L \tag{2}$$

The number density of the absorbing species, n_{abs}, can be expressed in terms of the absorption coefficient $k(\nu)$:

$$k(\nu) = S(T)\cdot\Phi(\nu-\nu_0)\cdot n_{abs} \tag{3}$$

$$\int_{-\infty}^{+\infty}\Phi(\nu-\nu_0)d\nu = 1 \tag{4}$$

Here, $S(T)$ is the spectral line strength, and $\Phi(\nu-\nu_0)$ is the normalized absorption profile. Using these relationships, the number density of the absorbing species can be derived as:

$$n_{abs} = -1\Big/S(T)\cdot L\int\ln[I(\nu)\Big/I0(\nu)]d\nu \tag{5}$$

Based on n_{abs} the volume mixing ratio (c_{abs}), which represents the concentration of a specific gas within the atmosphere or a gas mixture, can be expressed as:

$$c_{abs} = \frac{p_{abs}}{p} = \frac{n_{abs}\cdot k_B\cdot T}{p} \tag{6}$$

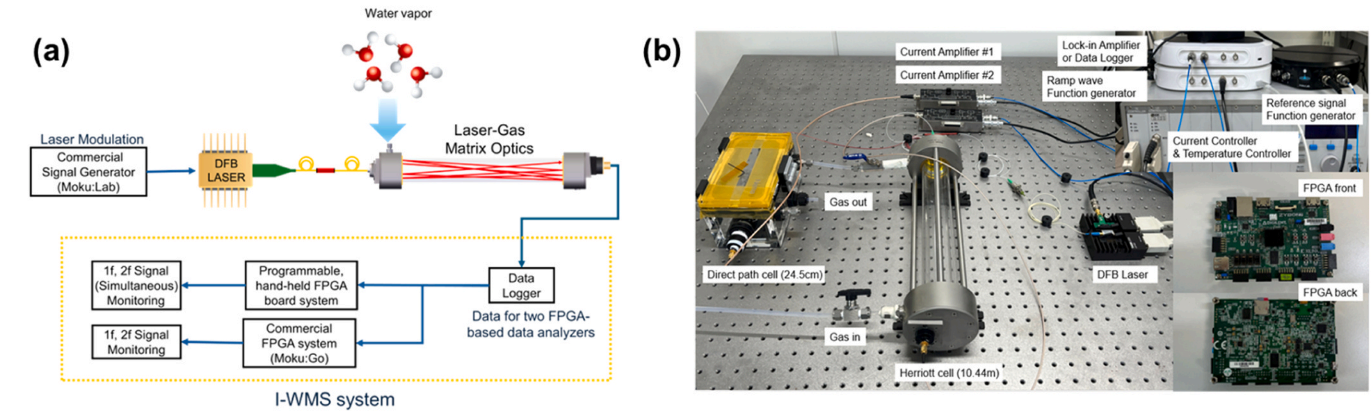

Fig. 1. Concept of the I-WMS system; (a) Schematic of the I-WMS; (b) Photograph of I-WMS.

where p is the total pressure of the gas mixture, p_{abs} is the partial pressure of the specific gas, k_B is the Boltzmann constant, and T is the temperature. Based on these equations, a TDLAS system can be constructed to measure the gas concentration by converting the absorption of the laser signal into concentration values.

WMS is a method for measuring sensitive absorption spectra. By applying sinusoidal modulation to the current of a laser, harmonic signals related to concentration can be measured, making it applicable for concentration monitoring [37]. Particularly, WMS effectively suppresses noise, such as 1/f noise, enabling improved sensitivity, precision, and signal-to-noise ratio (SNR) in trace gas measurements and harsh environments. To measure concentration using TDLAS-WMS, the laser current applied to a DFB laser is modulated by superimposing a sinusoidal modulation onto a ramp waveform [37–39]. The variation in laser current can be expressed by the following equation, where ν represents the central wavenumber, m is the parameter for sinusoidal modulation amplitude, and α denotes the sinusoidal modulation frequency.

$$I(\nu) = I(\nu + m\sin\alpha t) \tag{7}$$

If this equation is expanded using the Taylor series, it can be expressed as follows:

$$\begin{aligned} & I(\nu + m\sin\alpha t) \\ & = I(\nu) + (m\sin\alpha t)\frac{dI}{d\nu} + (\frac{m^2\sin^2\alpha t}{2!})\frac{d^2I}{d\nu^2} + \left(\frac{m^3\sin^3\alpha t}{3!}\right)\frac{d^3I}{d\nu^3} + \cdots \\ & = \left[I(\nu) + \frac{m^2}{4}\frac{d^2I}{d\nu^2} + \cdots\right] + (\sin\alpha t)\left[m\frac{dI}{d\nu} + \frac{m^2}{8}\frac{d^3I}{d\nu^3} \right. \\ & \left. + \cdots\right] + (\cos 2\alpha t)\left[-\frac{m^2}{4}\frac{d^2I}{d\nu^2} + \cdots\right] + \end{aligned} \tag{8}$$

Through this expansion, the sinusoidally modulated laser current can be expressed in terms of harmonic signals. The equation for the laser absorbance can be written as follows:

$$\tau(\nu) = \left(\frac{I_t}{I_0}\right)_\nu \tag{9}$$

Where $\tau(\nu)$ is the laser absorbance at given frequency, I_0 is the initial light intensity and I_t is the light intensity after the absorption.

From the above equation, the transmittance coefficient for the sinusoidally modulated current can be rewritten and expressed using a Fourier series as follows:

$$\tau(\nu_0 + a\cos(\omega t)) = \sum_{k=0}^{\infty} H_k(\nu_0, a)\cos(k\omega t) \tag{10}$$

$$H_0(\nu_0, a) = \frac{1}{2\pi}\int_{-\pi}^{+\pi} \tau(\nu_0 + a\cos\theta)d\theta \tag{11}$$

$$H_k(\nu_0, a) = \frac{1}{\pi}\int_{-\pi}^{+\pi} \tau(\nu_0 + a\cos\theta)\cdot\cos k\theta d\theta \tag{12}$$

$\tau(\nu_0 + a\cos(\omega t))$ is expressed as Fourier series, representing the transmittance coefficient as a sum of harmonic components. The Fourier coefficients that constitute this series are $H_k(\nu_0, a)$, representing the amplitude of k-th harmonic component in the modulated transmittance signal.

Here, the absorption coefficient can be expressed as follows:

$$k_\nu = P\cdot X\cdot S(T)\cdot\varphi_\nu \tag{13}$$

Given that P represents the pressure during measurement, X denotes the concentration, $S(T)$ is the spectral line strength, and φ_ν is the line shape function, the transmittance $\tau(\nu)$ can be approximated under the assumption that the absorption coefficient is very small $k_\nu\cdot L < 1$ where L is the optical path length. In this case, the transmittance can be expressed as:

$$\tau(\nu) = \exp(-k_\nu\cdot L) \approx 1 - k_\nu\cdot L = 1 - P\cdot X_i\cdot S\cdot\varphi(\nu)\cdot L \tag{14}$$

This approximation is valid for weak absorption conditions, allowing a linearized interpretation of the absorption signal in WMS.

For the second-harmonic 2 f signal in WMS, the detected signal can be derived by extracting the second harmonic component from the modulated transmission signal. Given the weak absorption approximation:

$$H_2(\bar{\nu}, a) = -\frac{S\cdot P\cdot X\cdot L}{\pi}\int_{-\pi}^{+\pi} \varphi(\bar{\nu} + a\cos\theta)\cdot\cos(2\theta)d\theta \tag{15}$$

$$X = -\frac{\pi\cdot H_2(\bar{\nu}, a)}{S\cdot P\cdot L\cdot\int_{-\pi}^{+\pi}\varphi(\bar{\nu} + a\cos\theta)\cdot\cos(2\theta)d\theta} \tag{16}$$

Through this, it can be derived that the height of the 2 f signal is directly related to the gas concentration. Therefore, the peak value of the 2 f signal is proportional to the concentration variation. This 2 f signal can be measured using a LIA, which extracts the desired harmonic signal from a noisy background by PSD. The LIA effectively isolates the 2 f component, allowing for precise and high-sensitivity gas concentration measurements in TDLAS-WMS [11], [40], [41].

2.2. Lock-in amplifier and digital lock-in amplifier in TDLAS-WMS

In lock-in detection, a reference signal with the same frequency as the measured signal is required. This reference signal can either be generated within the LIA or provided as an external input. The PSD determines the phase difference between the input and reference signals, adjusting the phase to ensure synchronization [42]. Its core principle is to mix and demodulate the input signal with a synchronous reference signal, producing in-phase or out-of-phase components [43]. In such applications, the LIA enables precise phase measurements, which are crucial for accurate displacement measurements. In an FPGA-based design, an LIA can estimate phase with a resolution limited to one period of the reference signal oscillator. One key advantage of FPGA-based designs is that, with a fixed system clock, the dynamic range of the digital-to-time converter only needs to account for the residual phase error within a single output clock cycle [44]. This FPGA-implemented LIA is referred to as a DLIA. A DLIA offers superior accuracy compared to analog LIA, benefiting from enhanced frequency synthesis, improved noise rejection, and more precise PSD [45–47]. Thus, DLIA is capable of accurately extracting both frequency and phase to amplify the desired signal, making it widely used in fields such as optics, photonics, nanotechnology, materials science, and sensor technology. This digital implementation provides several advantages over traditional analog LIA, including greater flexibility and faster signal processing speeds, ensuring superior performance. The ability of a DLIA to perform high-speed and highly accurate signal extraction and noise rejection makes it an essential tool in modern research and industrial applications where precision measurements are critical. Due to these advantages, FPGA-based DLIA have gained significant attention. FPGA technology enables the implementation of complex systems in a low-cost and compact form while maintaining high reliability and accuracy. This makes FPGA an ideal platform for designing DLIA systems, which is why FPGA-based DLIA designs are actively being researched [48].

DLIA operates using two inputs: the input signal and the reference signal. When these signals share the same frequency and phase, they are fed into the mixer (in other words, PSD) where they are multiplied, producing two components: a DC component (zero frequency) and an AC component (twice the frequency). Low-pass filter then removes the high-frequency component, leaving only the DC component, which corresponds to the desired signal. This selective filtering enables precise measurement of harmonic signals in TDLAS-WMS, where the reference signal frequency is adjusted to isolate specific harmonic components. By effectively isolating and extracting harmonic signals even in noisy conditions, the DLIA plays a crucial role in accurate gas concentration measurements, particularly in trace gas analysis. The 1 f and 2 f signals extracted by the DLIA are directly proportional to gas concentration. The 1 f signal primarily monitors laser power drift, helping minimize its effects and improving sensor accuracy. Specifically, the 1 f mean value represents the average power of the laser. By normalizing the max(2 f)/mean(1 f) ratio, a standardized signal, WMS-2 f/1 f, is obtained and expressed as the H_2O signal. This method allows for the compensation of laser power fluctuations, ensuring stable and reliable trace gas concentration measurements, such as those required for water vapour detection.

The performance of a DLIA is greatly affected by the characteristics of the filter, as the filter not only determines the output quality of the PSD but also influences the overall signal accuracy and noise rejection capability. The choice of filter type and parameters, particularly the cut-off frequency (f_c), plays a critical role in balancing signal accuracy and noise suppression. In Section 3.2, the detailed settings of cut-off frequency and the low-pass filter used in our CFTWH and PFTWH are described.

3. Experimental method

3.1. Experimental setup

As illustrated in Fig. 2, we used the same FPGA board for both (1) CFTWH and (2) PFTWH within the I-WMS system, and evaluated their performance with respect to dTDLAS. dTDLAS modulates the laser by linearly varying the current, enabling the direct evaluation of the absorption profile [49]. This method has the advantage of calibration-free characterization by utilizing pre-known spectral line parameters (spectral line intensity, shape function of absorption spectrum, etc.), meaning transmission losses caused by the reflectivity of mirrors or cloud droplets can be corrected without a calibration process, thus, proper to be used as a referencing measurement.

In this work, we propose a method for measuring water vapour concentration using an I-WMS by determining the 2 f/1 f signal ratio at the wavelength band corresponding to a water vapour absorption feature near 1370 nm. To enable parallel implementation of dTDLAS and I-WMS under identical conditions, two DFB lasers operating in the same spectral region were employed.

The I-WMS system consists of a commercial signal generator (Moku:Lab), two FPGA-based data analyzers (Moku:Go), a programmable, hand-held FPGA-based board system, and laser-gas matrix optics. While 1 f and 2 f signals are obtained from CFTWH and PFTWH, full absorption spectra from dTDLAS serve as a reference for evaluating their performance. Generally, dTDLAS is employed for high-concentration measurements, whereas previous studies on TDLAS-WMS have focused on low concentrations, typically ranging from 0 ppm to a few hundred ppm [38], [50,51]. In particular, TDLAS-WMS using DLIA has also been limited to this low concentration range [52,53]. In this study, we extended the concentration range and validated the reliability of water vapour measurements by performing linear regression analysis of TDLAS-WMS data against dTDLAS reference values.

The commercial DLIA in CFTWH has fixed input and output ports, making customization difficult. Since the experiment requires simultaneous measurement of 1 f and 2 f signals, but the DLIA in CFTWH processes them sequentially, it limited data processing speed. To overcome these limitations, a PFTWH system was developed as illustrated in Fig. 3. In PFTWH system, water vapour concentration measurements are performed using the WMS method to obtain 2 f/1 f signals. To achieve this, a DLIA was implemented on Xilinx FPGA (XC7Z020–1CLG400C) to process 1 f and 2 f signals in parallel.

As shown in Fig. 3, both systems employ an FPGA-based DLIA to extract the 1 f and 2 f signals. However, unlike the CFTWH in Fig. 3(a), the proposed PFTWH in Fig. 3(b) is implemented on a hand-held FPGA board with a custom DLIA design optimized for experimental conditions. In the CFTWH, the DLIA processes 1 f and 2 f signals separately in a serial manner, whereas in the PFTWH, it is designed to process 1 f and 2 f signals in parallel, reducing data processing time by half. Its flexible interface design allows for easy expansion and adaptation to various future applications. Moreover, the PFTWH offers significant advantages in terms of power consumption and installation convenience. It operates While the CFTWH requires an external power supply, the PFTWH operates solely via a USB connection, enabling experiments without additional equipment and enhancing mobility [54]. Since it can function on USB power, it can be easily applied to battery-powered environments, such as unmanned aerial vehicle.

The data processing method for dTDLAS in an optically thin environment was applied in the same way as in the previous studies on dTDLAS [55], further supporting the reliability of the data analysis. The measurement results using a commercial DLIA in a short optical path length (0.245 m) direct path cell with WMS were compared with the measurement results in a Herriott cell (10.4 m) using dTDLAS. For signal generation, DDS source function in the commercial analyzer (Moku:Go) was used to test I-WMS system and dTDLAS. Signal generator I generated the ramp wave applied to the DFB laser (NEL, NLK1E5DAAA) at

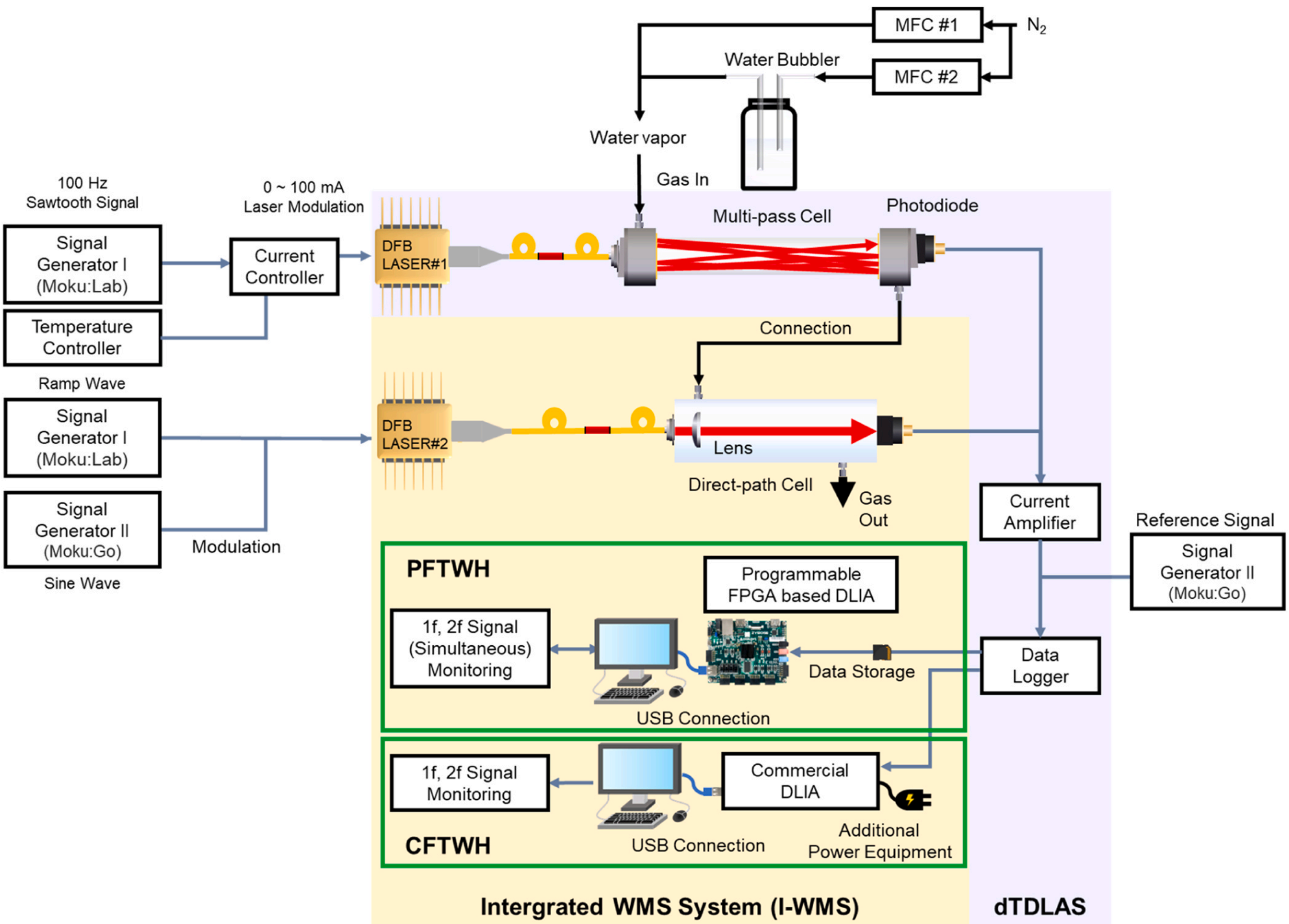

Fig. 2. Schematic of experimental setup; (yellow area) I-WMS system; (purple area) dTDLAS system.

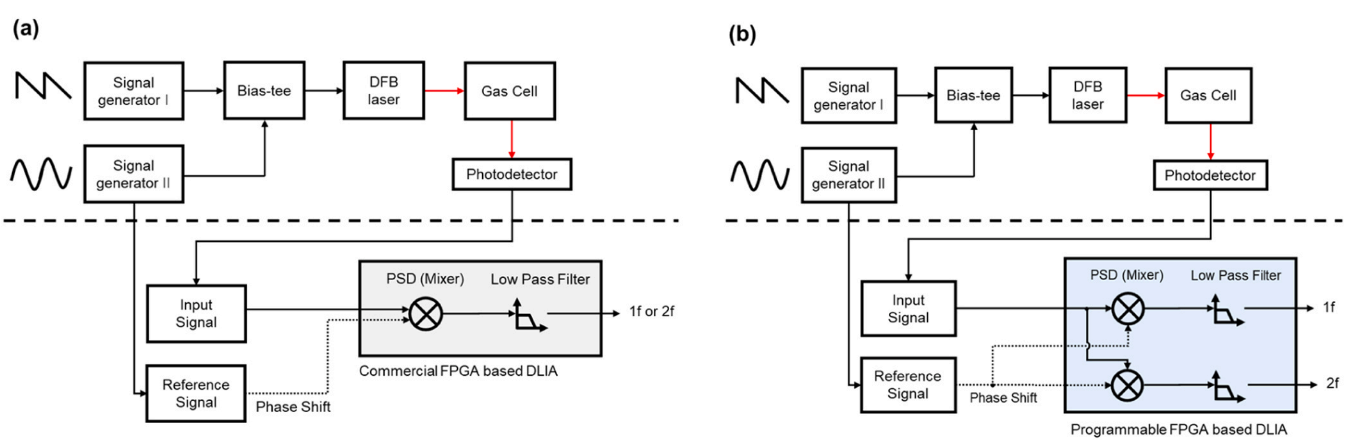

Fig. 3. Comparison of WMS configurations; (a) CFTWH system; (b) PFTWH system.

100 Hz and outputs it as a signal. Signal generator II (Moku:Lab) output a sine wave to modulate the ramp wave at 25 kHz and also output a 1 f reference signal at 25 kHz and a 2 f reference signal at 50 kHz with the same phase.

A Bias-Tee combined the triangle and sine waves and applied them to the DFB laser. The DFB laser emits 1370 nm light and was driven by a current controller (LDC8002) with a triangular waveform current and a temperature controller (TED8020) for temperature stabilization. The injected current modulated the wavelength by 1.75 cm^{-1} , covering the selected spectral line of the water vapour absorption spectrum. The light from the same DFB laser was emitted into two types of gas cells using a fiber-optic splitter (TW1300R5A1, THORLABS) connected to two fiber port collimators (PAF2-A4C, THORLABS). In developing TDLAS-WMS system, optimizing the sine wave modulation parameters enhances system performance by improving measurement accuracy and sensitivity.

A sine wave modulation signal was applied to the Bias-Tee of the laser mount, confirming the triangle wave modulation. The ramp and sine waves synthesized at the Bias-Tee were then applied to the laser. A parametric study was conducted to determine the optimal modulation frequency and amplitude. The applied ramp wave had a frequency of 10 Hz, with a minimum value of 500 mV and a maximum of 4.5 V. For optimization, sine waves with frequencies ranging from 10 kHz to 45 kHz and amplitudes ranging from 500 mV to 4 V were applied as modulation signals, and the corresponding 2 f signal was observed through the LIA. The optimal operating condition (2 V, 25 kHz) was determined as the one under which the 2 f signal exhibited the most symmetric signal profile.

For laser-gas matrix optics, water vapour was supplied to the gas cell, where the laser reflects inside, absorption occurs, and the absorption spectrum was measured by the photodetector. As in the previous study [55], a 24.5 cm short path cell was created, and water vapour concentration for both short and long paths was measured simultaneously using a single vapour source. Two mass flow controllers were used to establish the measurement environment, with one supplying dry nitrogen gas and the other supplying humid gas through a bubbler containing deionized water. By adjusting the flow rates of each mass flow controller, the concentration of water vapour supplied to the gas cell was varied. The mixed gas first enters the Herriott cell before passing into the single-path gas cell. The output laser then traveled through the gas cell and was detected by a photodetector. The transmitted light intensity was detected using an InGaAs photodiode (SM05PD5A) and amplified by a current amplifier (DLPCA-200).

The detected signals were collected through a data logger at a sampling rate of 500 kS/s. The raw absorption spectrum, which varies with water vapour concentration, was processed in the DLIA, where it is combined with the reference signal and filtered. The extracted 1 f and 2 f signals were then correlated with the water vapour concentration. While sharing DDS functions, the acquired signals after laser-gas matrix optics were processed in parallel through I-WMS and dTDLAS. CFTWH signals were processed using a commercial DLIA in Moku:Go. PFTWH signals were processed using a custom FPGA-based DLIA on a Xilinx, and dTDLAS followed the same processing method as in the previous studies [55].

3.2. Design of digital lock-in amplifier

In a TDLAS-WMS system, the DLIA's cut-off frequency must be carefully selected based on the modulation frequency to ensure stable acquisition of the 1 f and 2 f signals for water vapour concentration measurement. In this experiment, the modulation frequency was set to 25 kHz for symmetric signal. The reference signal that applies to the DLIA for 1 f detection was also set to 25 kHz, matching the modulation frequency, while for 2 f detection, the reference frequency was set to 50 kHz, twice the modulation frequency. This configuration ensures that mixing with the reference signal brings the desired signal near DC for stable detection.

Since the target signal is located near DC with a finite bandwidth, selecting an appropriate cut-off frequency is crucial. If the cutoff frequency is too low, the response time slows, potentially attenuating parts of the desired signal. Conversely, if it is too high, excessive noise may pass through, degrading the SNR. Experimental results confirmed that the bandwidth of both the 1 f and 2 f signals extends from tens to hundreds of Hz near DC, indicating that an optimal cut-off frequency should be set within the 100–500 Hz range.

The specific filter type used in the DLIA of the CFTWH is not publicly disclosed, but it allows adjustments to cut-off frequency and attenuation rate. In this experiment, the CFTWH's LIA filter was set to a cut-off frequency of 300 Hz with an attenuation rate of 12 dB, and its performance was compared to the PFTWH. For the FPGA-based DLIA in the PFTWH, a finite impulse response (FIR) filter with a windowed-Hamming design was implemented, with a cut-off frequency of 100 Hz and a 200-tap filter coefficient. The difference in cut-off frequency settings between the two systems was due to variations in filter architecture, and these differences were accounted to ensure that both systems operated under the most comparable conditions.

The order of the FIR filter introduces a group delay, calculated as (N − 1) / (2 × Fs) seconds [56]. For the 200-tap FIR filter at a 500 kHz sampling frequency, this corresponds to a delay of approximately 199 µs. This delay was explicitly considered during the DLIA design and parameter setting. Importantly, because FIR filters have a linear-phase characteristic, all frequency components are delayed by the same amount, ensuring that the delay itself does not distort high-frequency signal modulation.

3.3. Estimation of SNR, precision, and accuracy of CFTWH and PFTWH

To estimate 2 f signals into TDLAS-WMS system, both signals from raw detector and reference signals were processed using Python. The signal was demodulated via lock-in mixing, followed by a moving average filter to suppress high-frequency noise. A Savitzky-Golay filter was then applied to smooth the signal while preserving the shape of the harmonic signal [57]. The resulting simulated 2 f waveform can be quantitatively compared with experimental signals such as CFTWH and PFTWH to evaluate system performance. SNR was calculated as the ratio of the 2 f peak amplitude to the standard deviation of the residual, where the residual is defined as the pointwise difference between the simulated and experimental signals.

$$SNR = \frac{(\mathit{Peak\ of\ 2f\ signal})}{(\mathit{Standard\ deviation\ of\ residual})} \tag{17}$$

The 1 f and 2 f signals measured by the CFTWH and PFTWH were converted into concentration values using a linear regression equation. First, the accuracy between the water vapour concentrations obtained from the dTDLAS and the CFTWH was evaluated. Here, the dTDLAS-based concentration was used as the reference, and linear regression was applied to the WMS signals. The difference between the dTDLAS and CFTWH concentrations was used as a measure of accuracy, expressed as a percentage using the formula.

$$Accuracy = \left(\frac{Difference}{Reference}\right)\cdot 100\ (\%) \tag{18}$$

Subsequently, the accuracy between the PFTWH and CFTWH was calculated using the same method. In this case, the CFTWH concentration was used as the reference to evaluate the performance of the DLIA system, and the error rate was additionally presented.

Finally, the precision of long-term measurements using the PFTWH was evaluated. The average and standard deviation of the measured concentrations over 1000 s were calculated, and the precision was expressed as a percentage using the formula:

$$Precision = \left(\frac{\mathit{Standard\ deviation}}{Average}\right)\cdot 100(\%) \tag{19}$$

4. Results and analysis

4.1. Summary of PFTWH development

The commercial DLIA in the CFTWH functions as a conventional LIA but is limited by fixed input and output ports, making customization difficult. Additionally, since it cannot simultaneously process 1 f and 2 f signals, sequential signal processing was required, limiting the system's speed. To address these limitations, we developed a PFTWH. Like the CFTWH, the PFTWH digitally extracts 1 f and 2 f signals using an FPGA-based DLIA. However, the PFTWH is implemented on a hand-held Xilinx FPGA with a custom DLIA design optimized for the experimental environment. Since the DLIA is designed according to the board's specifications, resource limitations may impact performance. Unlike an analog filter in an LIA, digital processing may still allow some high-frequency components or noise. However, in WMS, water vapour concentration is determined by extracting the mean value of the 1 f signal and the peak value of the 2 f signal, meaning noise in the side regions does not affect concentration measurement.

As demonstrated in Fig. 3, the FPGA-based DLIA was designed to process the essential 1 f and 2 f signals in parallel within the WMS method, reducing the data-processing time by half compared with the conventional system. Moreover, its flexible interface allows easy expansion and adaptation for various future applications. In addition, the PFTWH offers advantages in terms of power efficiency and ease of installation, with its USB-powered operation making it well-suitable for battery-powered environments. In I-WMS platform, while both the CFTWH and PFTWH measure water vapour concentration using the 2 f/1 f harmonic ratio, dTDLAS determines concentration through direct absorption spectroscopy, where the absolute absorption signal is measured and concentration is calculated using Beer-Lambert's law. The consumption of FPGA Resources in PFTWH is summarized as Table 1.

Generally, digital signal processing (DSP) blocks enable faster computation compared to using look-up tables (LUTs) [58]. As shown in Table 1, the FPGA in this system utilizes 218 out of 220 DSP blocks (99.1 %) and 128 out of 140 BRAM blocks (91.4 %), optimizing the system within the board's specifications. Although the DSP blocks utilization is very high, this design choice reflects the active use of DSP block resources to maximize DLIA performance and does not impose limitations on the intended operation of the system. The FPGA operates at a 100 MHz clock speed and communicates with the computer via the AXI4-Lite protocol. Data related to the absorption spectrum and the waveform generator is stored on an SD card and transmitted to the FPGA. With a 500 kHz sampling frequency, the FPGA sequentially processes the data over time to output the 1 f and 2 f signals, ensuring efficient data handling and high-performance signal processing. In the PFTWH design, the 100 MHz FPGA clock provides approximately 200 clock cycles for processing each 500 kHz data sample, offering sufficient timing margin to reliably perform all computations and ensure stable operation.

4.2. Comparison of CFTWH with dTDLAS

Using a gas cell with a long optical path length improves measurement sensitivity, as increased absorption occurs even at low concentrations.

Since the optically thick condition does not occur in the short path cell, the focus was on determining the lower concentration limit. In the CFTWH system, as shown in Fig. 4(a), 2 f signal data was measured using the WMS method in the low concentration range and analyzed using the dTDLAS method. Below 70 ppm, the waveform shape deviated from the patterns observed at higher concentrations. Based on this observation, the lower concentration limit was set to 70 ppm.

However, at higher concentrations, excessive light absorption can cause a central region in the absorption spectrum where the signal approaches 0 V, a condition known as the optically thick state. In this state, it becomes difficult to calculate concentration based on the absorption spectrum area. Due to the long optical path length of the Herriott cell used in this experiment, optically thick regions appear at high concentrations, as shown in Fig. 4(b). The optically thick condition usually refers to a circumstance in which high absorbance caused by high absorber density, high spectral line intensity, and long absorption path

Table 1
FPGA resource consumption in PFTWH.

Resource	Available	Utilization
LUTs	53,200	20,814 (39.1 %)
LUTRAM	17,400	154 (0.900 %)
F/F	106,400	23,164 (21.8 %)
BRAM	140	128 (91.4 %)
DSP blocks	220	218 (99.1 %)

length interferes with the precise identification of the gas concentrations of interest. The upper and lower measurement limits were determined for both the short path cell and the Herriott cell. Fig. 4(b) highlights the region in the dTDLAS system where the optically thick condition does not occur. Based on this, the upper concentration limit was set to 5800 ppm, ensuring reliable measurement conditions.

We selected the 70–5800 ppm range as it represents the window where sufficient SNR can be obtained at low concentrations while maintaining linearity without optical saturation at high concentrations. The experiment was conducted within this range. Previous WMS-based gas concentration studies have primarily focused on low concentrations, typically between 0 ppm and a few hundred ppm. Specifically, to detect trace gases at levels spanning ppb to ppm, researchers have leveraged calibration-free WMS techniques using quantum cascade lasers to secure stable sensitivity even at extremely low concentrations [37]. Furthermore, diverse strategies have been implemented to enhance measurement precision. most notably through the application of the 2 f/1 f harmonic ratio method, which effectively addresses challenges such as modulation nonlinearity and background noise [38]. In addition, multi-pass optical designs, exemplified by the use of Herriott cells, have been actively adopted to minimize optical path loss and consequently improve measurement reliability [59]. This study demonstrates the capability to measure water vapour concentrations over a broader range of 70 ppm to 5800 ppm, highlighting the potential of WMS technology for climate science applications.

Since the FPGA supports parallel computation at the hardware level, it can simultaneously process the 1 f and 2 f signals while performing high-speed calculations. Additionally, the FPGA-based DLIA offers high circuit integration density and low power consumption, enabling device miniaturization and improved mobility. The CFTWH utilizes an FPGA-based DLIA within commercial hardware to measure water vapour concentration using the TDLAS-WMS method, which relies on the 2 f/1 f harmonic ratio. In contrast, dTDLAS directly measures the absorption signal and calculates concentration using the Beer-Lambert law. While dTDLAS provides highly accurate water vapour concentration measurements, it requires complex data processing, including non-linear spectrum fitting and multi-variable calibration. On the other hand, the CFTWH system offers simpler data processing and faster concentration measurement but requires calibration using a reference. To evaluate the CFTWH's performance against dTDLAS, a linear regression analysis was performed, comparing the CFTWH results with concentrations obtained from dTDLAS. The applicable concentration range for both methods was identified as 70 ppm to 5800 ppm, as the prior study confirmed that this range meets the optically thin condition in the same optical setup. The same dTDLAS data processing method as in the previous study was applied in this optically thin environment, further validating the reliability of the data analysis. Within this concentration range, as shown in Fig. 5, measurements from the CFTWH system were compared to those from a Herriott cell (10.4 m) using dTDLAS.

The 1 f and 2 f signals collected from the CFTWH showed a proportional relationship with concentration. Fig. 5(a) presents the 1 f signal variation as a function of concentration, showing that the waveform changes with increasing concentration. The mean value of the 1 f signal represents the laser's average power and provides insights into power drift phenomena. Fig. 5(b) illustrates the 2 f signal variation with concentration, confirming that the peak value of the 2 f signal increases as concentration rises. This peak value directly correlates with water vapour concentration. Fig. 5(c) presents a linear regression analysis of the CFTWH data as a function of concentration. The 1 f signal helps monitor power drift in the laser, minimizing its effects and improving sensor accuracy. By comparing max(2 f)/mean(1 f) values, a standardized WMS-2 f/1 f signal was obtained. This signal, denoted as the H_2O signal, was plotted against the water vapour concentration obtained from dTDLAS, revealing the following relationship:

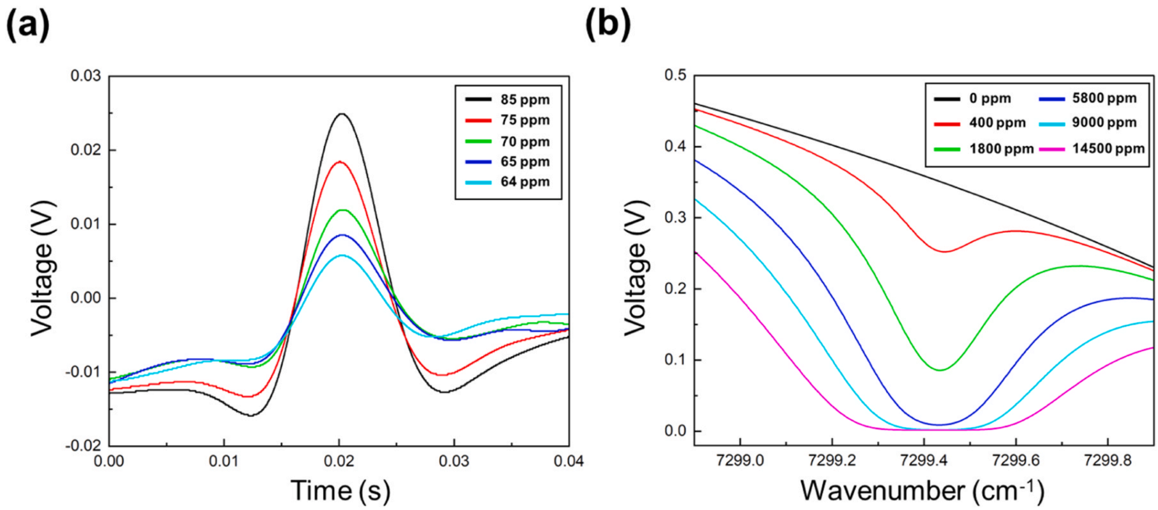

Fig. 4. Experimental Results for the Determination of Concentration Range; (a) Results of the lower limit of 2 f signal under low water vapour concentration environments; (b) Results of upper limit of dTDLAS under non-optically thick condition.

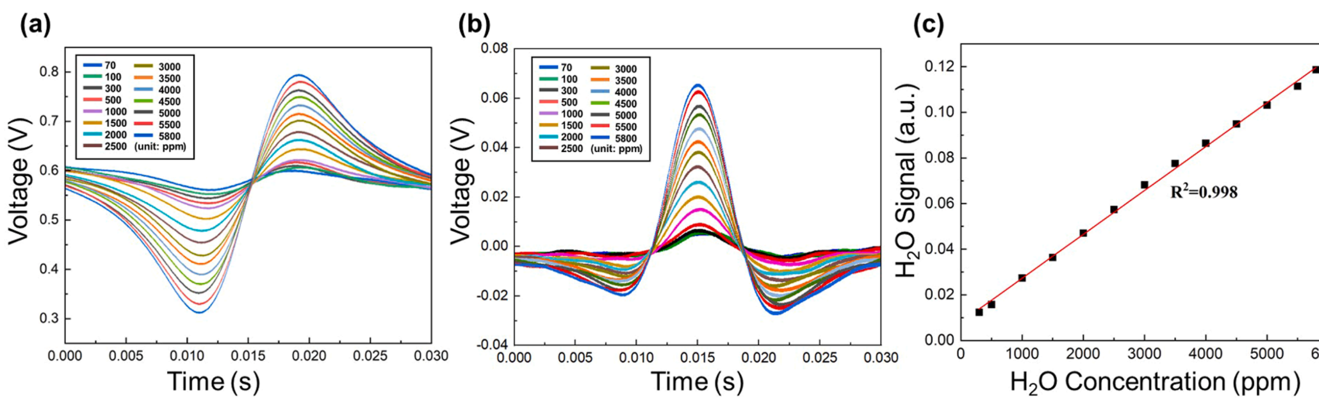

Fig. 5. Experimental results with respect to various concentrations in I-WMS; (a) Results of 1 f signal from 70 ppm to 5800 ppm in CFTWH; (b) Results of 2 f signal from 70 ppm to 5800 ppm in CFTWH; (c) Results of linear regression of H_2O signal (WMS-2 f/1 f) with respect to concentrations measured in dTDLAS.

$$H_2O\ Signal\left(WMS - \frac{2f}{1f}\right) = a\cdot(Water\ vapour\ concentration) + b \tag{20}$$

a is the slope of the linear regression line and b is the y-intercept of the linear regression line.

This equation models the linear relationship between the water vapour concentration and the WMS-2 f/1 f signal intensity.

$$y = 0.00001924x + 0.00803 \tag{21}$$

where y represents the H_2O signal and x represents the water vapour concentration. The coefficient of determination (R^2) was found to be 0.998, confirming a strong linear correlation between the WMS-2 f/1 f signal and water vapour concentration in the range of 70 ppm to 5800 ppm. The data related to the linear regression equation in Fig. 5(c) is presented in a Table 2, to verify the correlation between the CFTWH and dTDLAS concentrations.

A linear regression analysis was also conducted for the PFTWH to examine the relationship between the WMS-2 f/1 f signal and water vapour concentration. The following equation represents the linear regression result for the PFTWH. Similar to the CFTWH case, it demonstrates the linearity of the H_2O signal with respect to water vapour concentration.

$$y = 0.00002011x + 0.00867 \tag{22}$$

Here, y represents the H_2O signal expressed as WMS-2 f/1 f, and x corresponds to the water vapour concentration measured using dTDLAS.

Table 2
Accuracy analysis of CFTWH compared to dTDLAS.

dTDLAS Concentration [ppm]	CFTWH Concentration [ppm]	Accuracy [ppm]
300	225	75.0 (25.0 %)
500	402	98.0 (19.6 %)
1000	1010	10.0 (1.00 %)
1500	1470	30.0 (2.00 %)
2000	2030	30.0 (1.50 %)
2500	2570	70.0 (2.80 %)
3000	3130	130 (4.33 %)
3500	3620	120 (3.43 %)
4000	4080	80.0 (2.00 %)
4500	4510	10.0 (0.222 %)
5000	4940	60.0 (1.20 %)
5500	5370	130 (2.36 %)
5800	5750	50.0 (0.862 %)

The H_2O signal measured by the CFTWH was converted to ppm units and directly compared with the dTDLAS concentrations as shown in Table 2. In the low concentration range ($\leq$500 ppm), accuracy was relatively lower due to a reduced SNR, making the measurement more sensitive to minor signal variations. Beyond this range, accuracy remained within 1–4 %, confirming that the CFTWH delivers reliable measurement performance across the entire tested concentration range.

The linearity and accuracy of the dTDLAS confirms the system's measurement stability.

4.3. Comparison of PFTWH with CFTWH

Since the data processing methods of the TDLAS-WMS and dTDLAS systems differ, it is more appropriate to compare the PFTWH with the validated CFTWH. A direct comparison between dTDLAS and PFTWH would introduce two independent variables, specifically the TDLAS-WMS experimental setup and the DLIA performance, making it difficult to isolate the effect of each. Therefore, we first validated the measurement reliability of the WMS method using the CFTWH and then assessed the FPGA-based DLIA performance in the PFTWH through a step-by-step verification process. This approach ensured the reliability of the experimental results.

As demonstrated in the previous Section 4.2, the CFTWH exhibited high linearity, with a coefficient of determination (R^2) of 0.998 in the regression analysis of the 2 f/1 f signal, using dTDLAS as the reference for water vapour concentration. Since the PFTWH and CFTWH use the same signal processing method, direct validation is possible between the two systems, ensuring that the reliability of CFTWH is transferred to PFTWH. Like the CFTWH, the PFTWH collected data from a short optical path (24.5 cm) using WMS. The performance of PFTWH can be evaluated based on the 1 f and 2 f data collected from the CFTWH.

In this section, at given concentrations of 300 ppm and 5500 ppm within the range of 70–5800 ppm, the 1 f and 2 f signal measurements from the CFTWH and PFTWH were compared as demonstrated in Fig. 6. Fig. 6(a) presents the 1 f signals obtained from both systems at 300 ppm, while Fig. 6(b) shows the corresponding 2 f signals. In the low concentration range of 300 ppm, the 1 f and 2 f signals from the PFTWH closely match those from the CFTWH. Fig. 6(c) and Fig. 6(d) display the 1 f and 2 f signals, respectively, from both systems at 5500 ppm. Again, the PFTWH's measurements align well with the CFTWH. The results show that the PFTWH's measurements exhibit similar waveforms to those of the CFTWH. We established an environment to directly evaluate the 1 f and 2 f signal waveforms obtained from the PFTWH and CFTWH under identical concentration conditions, and the results indicate that both systems exhibit highly consistent waveforms with the same pattern and a high degree of alignment. Standard deviation of 1 f and 2 f signals, measured in voltage levels, were observed to be 0.002–0.003, and ~0.001, respectively, as presented in Table S1. This confirms that at given concentrations, the PFTWH achieves the same signal processing accuracy as the CFTWH while maintaining stable measurement performance through parallel processing with the FPGA-based DLIA.

Concentration measurements were conducted across a wide range from 70 to 5800 ppm using the PFTWH system as shown in Fig. 7.

The primary objective was to validate the linearity of the PFTWH's 1 f and 2 f signal measurements, confirming its reliability across different concentration levels. In the PFTWH, absorption spectrum data, along with the 1 f signal and its reference, are simultaneously recorded at a 500 kHz sampling rate. The system then processes both 1 f and 2 f signals in parallel, generating waveforms corresponding to concentration variations. Fig. 7(a) presents the 1 f signal of the PFTWH with respect to concentration. As the concentration increases, the peak-to-peak difference within one cycle of the 1 f signal becomes larger. The amplitude increases with concentration changes, while the mean value of the 1 f signal remains stable. This stability provides insight into laser power drift, indicating that despite fluctuations in concentration, the 1 f mean value remains constant. Fig. 7(b) illustrates the 2 f signal variation in response to concentration changes. The peak value of the 2 f signal,

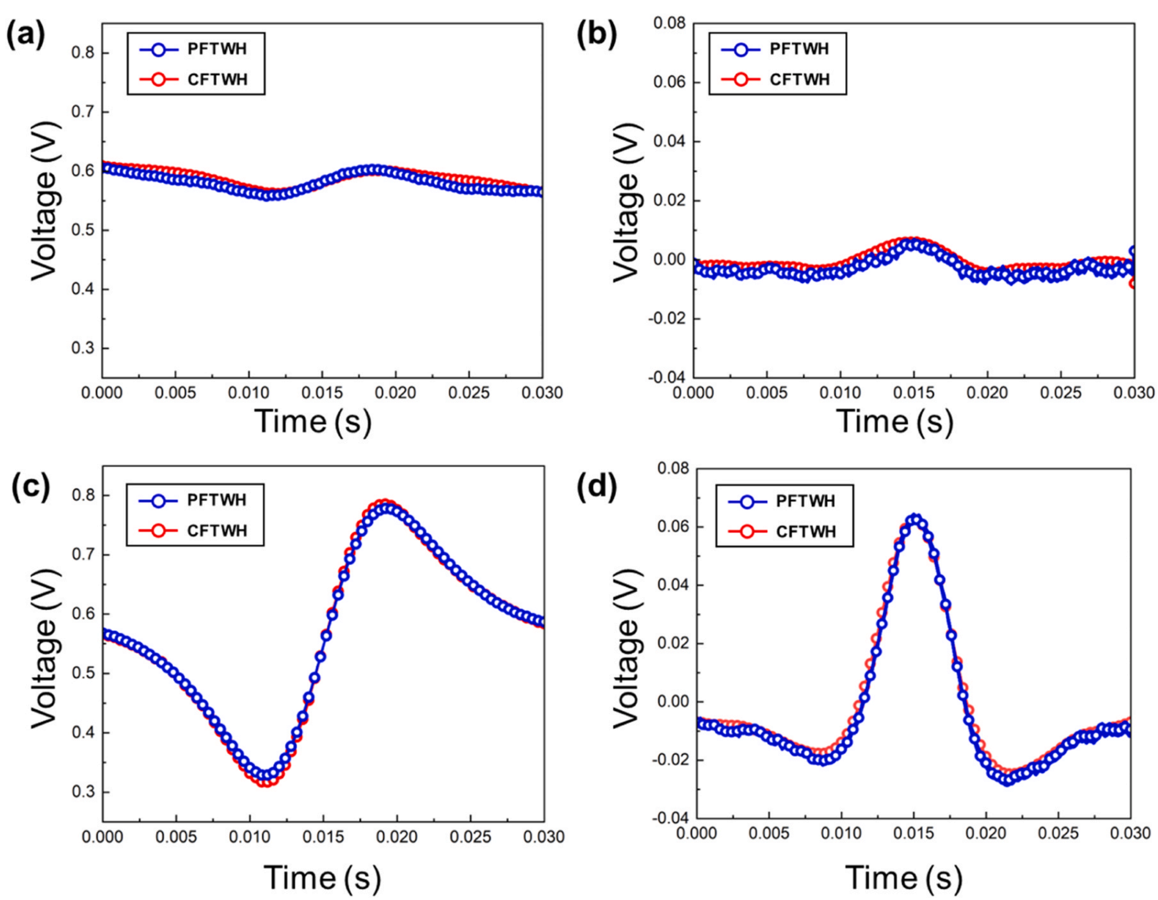

Fig. 6. Experimental results of CFTWH and PFTWH; (a) Results of 1 f signal between CFTWH and PFTWH at 300 ppm; (b) Results of 2 f signal between CFTWH and PFTWH at 300 ppm (c) Results of 1 f signal between CFTWH and PFTWH at 5500 ppm; (d) Results of 2 f signal between CFTWH and PFTWH at 5500 ppm.

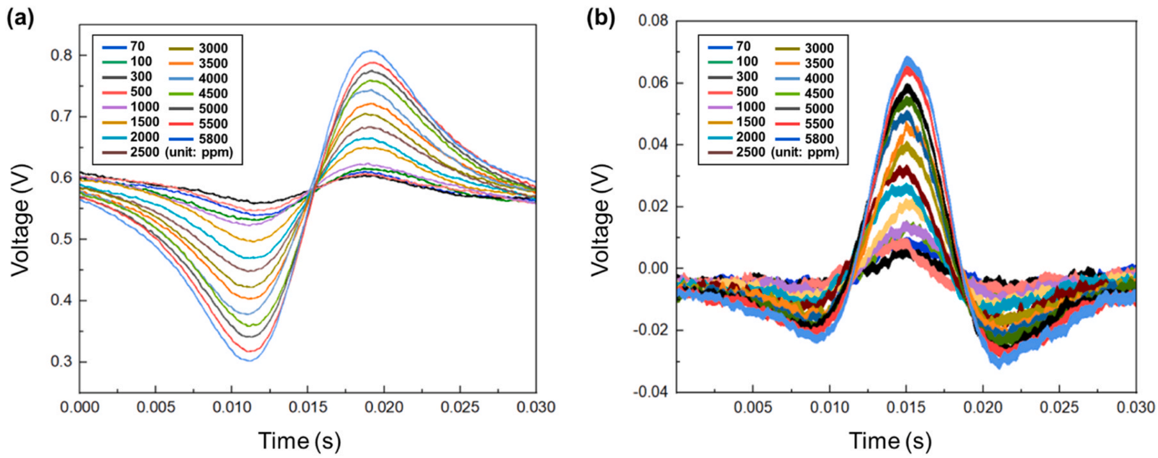

Fig. 7. Experimental results with respect to various concentration in I-WMS; (a) Results of 1 f signal from 70 to 5800 ppm in PFTWH; (b) Results of 2 f signal from 70 to 5800 ppm in PFTWH.

which is directly related to concentration, increases as concentration rises. By analyzing the 1 f and 2 f signals over the 70–5800 ppm range, the results confirm a proportional relationship with concentration, demonstrating the system's linearity in concentration measurement. The broader appearance of the 2 f signal waveform in Fig. 7(b) is due to the filter performance in the FPGA. The FPGA used in this PFTWH is the ZyboZ720, optimized for implementing DLIA according to the board specifications. The filter has reduced performance for noise below 100 Hz, impacting the 2 f signal, which has a higher reference frequency. Consequently, the 2 f signal waveform appears thicker. Despite this, the system accurately detects 2 f peak values across the 70–5800 ppm range. The data analysis for the PFTWH is shown in Fig. 8, comparing its performance with the CFTWH. This comparison highlights the reliability of the PFTWH in measuring concentrations within the same range as the CFTWH.

The results in Fig. 8 demonstrate that both systems produce consistent and accurate concentration measurements, confirming that the PFTWH maintains the measurement reliability of the CFTWH while

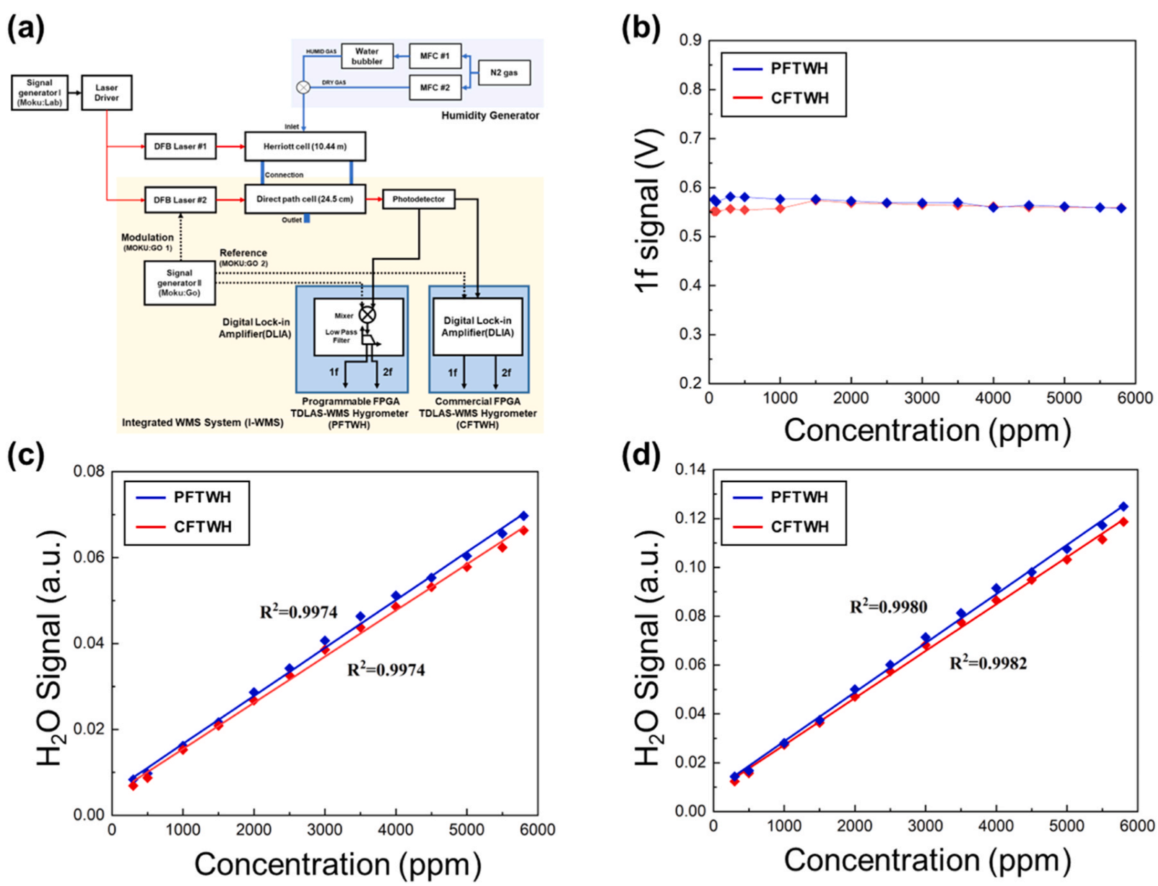

Fig. 8. Analysis of experimental data (Fig. 5 and Fig. 7) obtained by PFTWH and CFTWH with respect to various concentrations measured in dTDLAS; (a) Schematic of CFTWH and PFTWH; (b) Results of 1 f - WMS mean value; (c) Results of linear regression analysis of 2 f – WMS peak value; (d) Results of linear regression analysis of H_2O signal (WMS-2 f/1 f).

providing enhanced portability due to its hand-held FPGA-based design. Fig. 8(b) presents the 1 f mean values for both the PFTWH and CFTWH as a function of concentration. This comparison provides insights into how both systems respond to concentration variations, demonstrating consistency in 1 f mean values across both systems. Since the 1 f mean value reflects laser power drift, its similar behavior in both systems ensures accurate concentration measurements. For the 1 f signal, while the amplitude increases with concentration, the midpoint remains constant. Experimental results confirmed that the 1 f midpoint consistently averages 0.570. Additionally, when comparing the 1 f signal midpoints of CFTWH and PFTWH, the deviation was found to be less than 5 %, as tabulated in Table 3. This confirms that the PFTWH effectively processes the 1 f signal in response to concentration variations.

Fig. 8(c) presents the linear regression analysis of the 2 f peak values for both PFTWH and CFTWH as a function of concentration. Since the 2 f peak value is directly proportional to concentration, it increases linearly with increasing concentration. Both systems exhibit this trend consistently, and the linear regression analysis for 2 f peak yields a coefficient of determination (R^2) of 0.9974 for both, confirming a strong linear relationship in the data. Fig. 8(d) illustrates the WMS-max(2 f)/mean (1 f) signal variation with concentration for both PFTWH and CFTWH. As previously shown in Fig. 5(c), where the WMS-2 f/1 f signal is expressed as an H_2O signal through cross-validation between dTDLAS and the CFTWH, this figure follows the same methodology between PFTWH and CFTWH. The linear regression analysis reveals R^2 values of 0.9982 for CFTWH and 0.9980 for PFTWH, demonstrating the reliability and consistency of the PFTWH with respect to the CFTWH. As presented in Table 4, the accuracy of the PFTWH and CFTWH measured concentrations remains within 0.1–4 % across most of the tested range, demonstrating high precision and measurement reliability.

The experimental comparison between the two systems confirms that the PFTWH provides stable and accurate measurements with a level of linearity comparable to the CFTWH. The strong R^2 values (0.9982 for CFTWH and 0.9980 for PFTWH) validate that the PFTWH can be reliably used for water vapour concentration measurements across a broad range. These results further enhance the credibility of the PFTWH for practical applications in climate science, proving its ability to maintain high accuracy and performance.

4.4. Long - term stability analysis of PFTWH

To validate stability of PFTWH over extended periods, long-term measurement experiments were conducted. This characteristic significantly enhances the system's usability in environments where mobility is crucial. Demonstrating stability through long-term measurements is an essential step in ensuring the PFTWH's reliability in real-world applications, reinforcing confidence in its practical deployment.

Table 3
PFTWH - deviation with respect to CFTWH.

Mean Concentration [ppm]	PFTWH-1 f value [V]	1 f deviation (%)	PFTWH-2 f value [V]	2 f deviation (%)
70	0.576	4.39	0.00834	21.2
100	0.571	3.51	0.00975	11.6
300	0.582	4.50	0.0162	5.96
500	0.580	4.70	0.0216	3.59
1000	0.576	3.50	0.0287	7.14
1500	0.576	0.457	0.0342	4.97
2000	0.573	0.718	0.0407	5.45
2500	0.569	0.250	0.0463	5.87
3000	0.569	0.692	0.0512	5.33
3500	0.570	1.06	0.0553	4.05
4000	0.559	0.418	0.0604	4.41
4500	0.564	0.718	0.0656	5.18
5000	0.561	0.178	0.0697	5.13
5500	0.559	0.0180	0.00834	21.2
5800	0.558	0.177	0.00975	11.6

Table 4
H_2O signal accuracy and deviation rate between PFTWH and CFTWH system.

PFTWH Concentration [ppm]	CFTWH Concentration [ppm]	Accuracy [ppm (%)]	(2 f/1 f) deviation (%)
282	225	57.0 (25.3 %)	16.0
404	402	2.00 (0.498 %)	6.60
964	1010	46.0 (4.55 %)	2.48
1440	1470	30.0 (2.04 %)	3.13
2060	2030	30.0 (1.48 %)	6.38
2560	2570	10.0 (0.389 %)	4.72
3120	3130	10.0 (0.319 %)	4.72
3610	3620	10.0 (0.276 %)	4.77
4120	4080	40.0 (0.980 %)	5.77
4450	4510	60.0 (1.33 %)	3.30
4920	4940	20.0 (0.405 %)	4.23
5400	5370	30.0 (0.559 %)	5.20
5780	5750	30.0 (0.522 %)	5.30

As shown in Fig. 9, the measured data was analyzed using Allan deviation to determine the optimal integration time, to minimize local time drift effect.

The Allan deviation analysis of the 2060 ppm and 5220 ppm data confirmed that ~ 30 s is the optimal integration time to reduce time drift effect. Then, it was applied to the long-term measurement analysis. The plot also demonstrates the Allan deviation exhibits a τ (averaging time) $^{-0.5}$ dependence. Through this analysis, noise characteristics and long-term drift were identified and corrected. To assess the long-term measurement capability of the PFTWH using WMS, as demonstrated in Fig. 10, data was collected for 1 f and 2 f signals over 1000 s at ten different water vapour concentrations: 417 ppm, 1020 ppm, 1590 ppm, 2050 ppm, 2590 ppm, 3150 ppm, 3600 ppm, 4050 ppm, 4430 ppm, and 5170 ppm. The Allan deviation analysis was also extended to the low concentration regime. At 417 ppm, the analysis revealed Allan deviations of 52.1 ppm at $\tau = 0.1$ s, 4.26 ppm at $\tau = 30$ s (~1.02 % relative precision), and a minimum of 1.69 ppm at $\tau \approx 47.4$ s. These results demonstrate that the system maintains ~1 % precision with 30 s averaging even under low concentration conditions, which is consistent with the long-term stability of 2.40 % reported in Table 5. Overall, these findings confirm the robustness of the Allan deviation results across the entire concentration range, highlighting the reliability of the system's measurement performance.

The DLIA integrated within the PFTWH enabled simultaneous processing of 1 f and 2 f signals, and the measured concentrations derived from the absorption spectrum were plotted over time on Fig. 10. The results confirmed that the PFTWH could reliably measure water vapour concentration over 1000 s at a constant concentration, maintaining measurement stability throughout the experiment. Nevertheless, in the direct path cell, the influence of flow dynamics can be reflected in the measurements, with a modest influence observed at concentrations above 2000 ppm.

The PFTWH validated the capability of the WMS-based concentration measurement system to perform high-speed signal analysis and parallel processing of 1 f and 2 f signals. Additionally, with 30 s time averaging and stable operation over 1000 s measurement periods, the system proved to be well-suited for extended-duration experiments. Table 5 presents precision values calculated from the data in Fig. 10. The precision ranged from approximately 1–2 %, maintaining high consistency and stability over 1000 s across most concentration levels. At

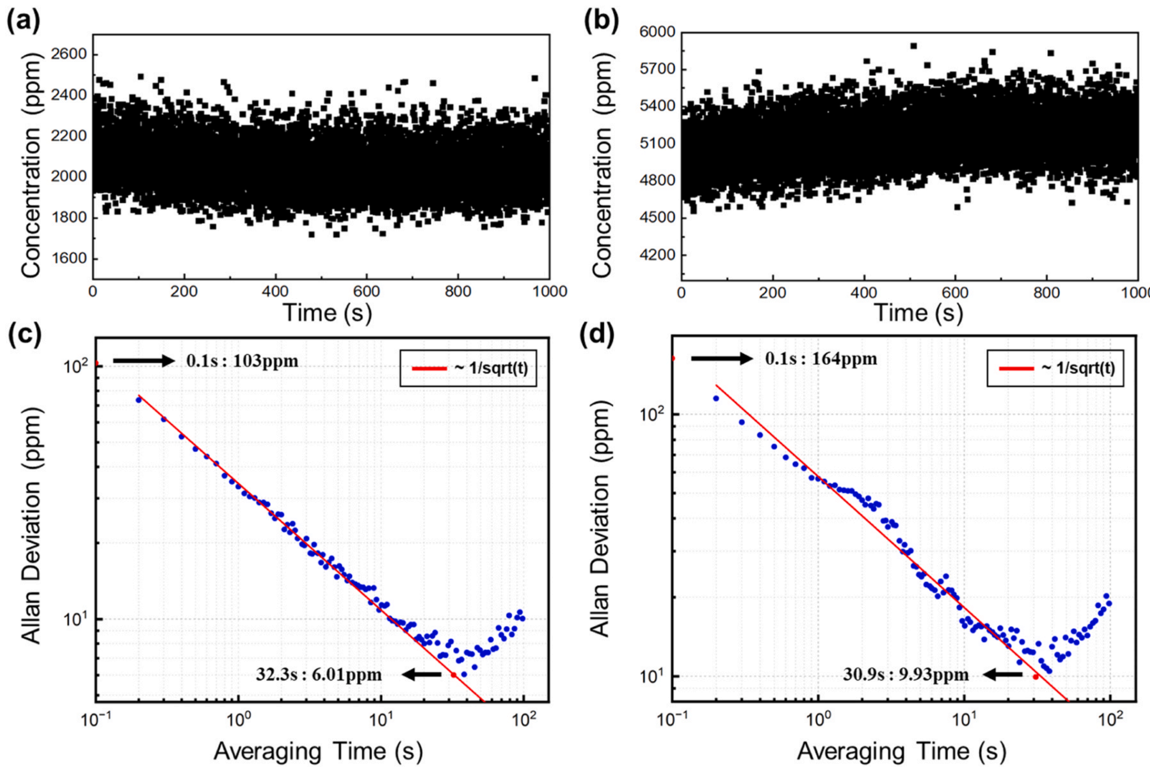

Fig. 9. (a), (b) Raw concentration data at 2060 ppm and 5220 ppm, respectively; (c), (d) Corresponding Allan deviation plots indicating white noise behavior, with red trend lines exhibiting a τ (averaging time)$^{-0.5}$ dependence.

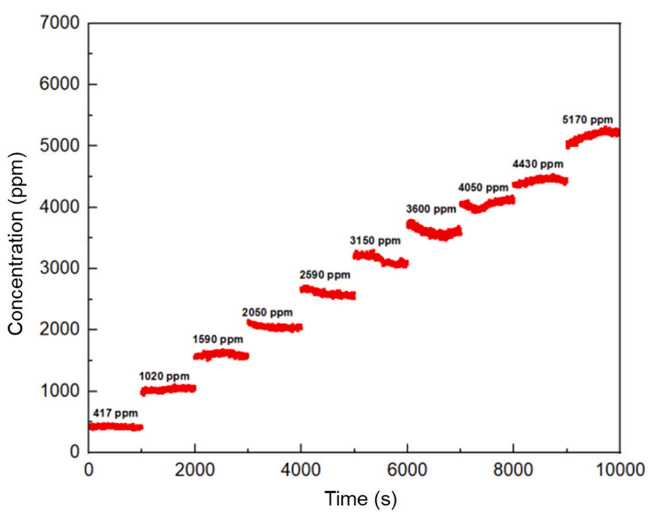

Fig. 10. Results of PFTWH in long - term measurements at various concentrations.

lower concentrations, precision values were relatively smaller, but they gradually decreased as concentration increased, resulting in more stable measurements at higher concentrations. The measured precision showed 2–5 times more improvement than the raw data, as presented in Table S4, validating that the optimal integration time we chose is proper. This confirms that PFTWH ensures reliable and consistent accuracy across the entire measurement range. To further assess its accuracy, the PFTWH measurements were also compared with reference values from dTDLAS.

As shown in Fig. 11(a), linear correlation ($R^2 = 0.997$) was obtained between the two systems, and Fig. 11(b) presents the corresponding residuals, calculated as the difference between the PFTWH and the reference, plotted below the regression curve to illustrate their distribution. To further verify the robustness of the regression, additional analyses were carried out by progressively excluding high-concentration points and by applying residual-based weighting. In both cases, the slope and intercept remained nearly identical to the original fit ($R^2 > 0.997$), demonstrating that the calibration is not significantly influenced by high-concentration end points or residual scatter. (Fig.S1, Fig.S2) The results confirm that the PFTWH maintains reliable accuracy across the entire tested concentration range, reinforcing its capability for stable and precise measurements.

To improve measurement accuracy under varying environmental conditions, interference suppression [60] and temperature/pressure compensation algorithms [61] can be integrated. This approach compensates for changes in molecular absorption line strength due to temperature variations, and corrects line broadening and line shifts induced by pressure fluctuations, thereby enhancing the reliability of the concentration inversion process.

Table 5
Long-Term precision of PFTWH over 1000 s for each concentration.

PFTWH Mean Concentration [ppm]	Precision [ppm (%)]
417	10.0 (2.40 %)
1020	18.2 (1.77 %)
1590	24.1 (1.51 %)
2050	21.6 (1.05 %)
2590	45.5 (1.76 %)
3150	59.6 (1.89 %)
3600	55.4 (1.54 %)
4050	52.4 (1.30 %)
4430	36.9 (0.830 %)
5170	69.5 (1.35 %)

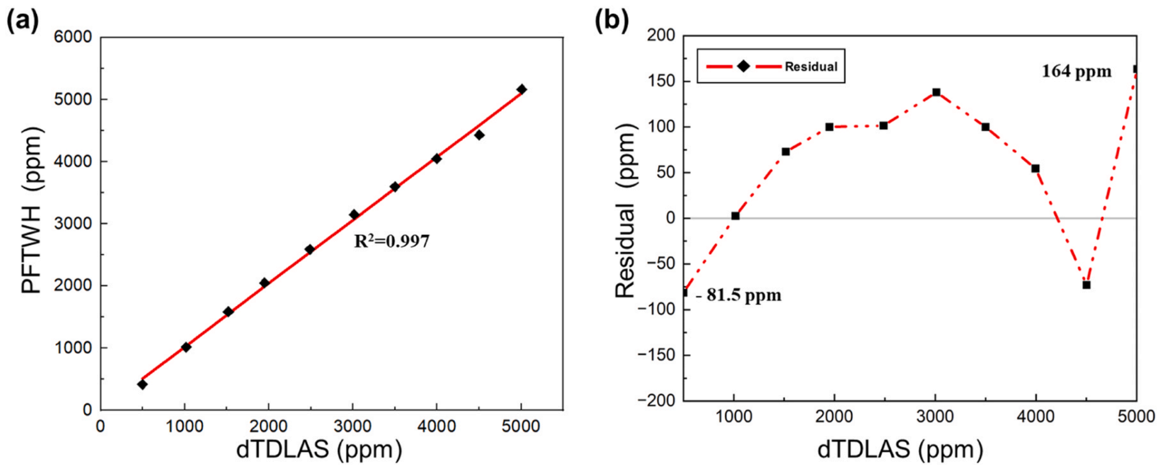

Fig. 11. Comparison of PFTWH and dTDLAS concentration results; (a) Linear relationship between H_2O concentrations measured by both methods (R^2 = 0.997); (b) Residuals relative to dTDLAS, ranging from −81.5 ppm to + 164 ppm across the tested range.

4.5. Atmospheric implications

To be able to envision applicability of PFTWH to atmospheric processes, there are a number of issues that PFTWH may face when implemented in airborne measurement such as sampling, low pressure measurement, accuracy and precision, and multiplex detection capability. Additionally, this study considered the impact of uncertainties arising from the generation of water vapour concentration on the overall measurement accuracy. For this purpose, we referred to the uncertainty analysis performed for the integrative methods and processing methods for hygrometer (IMPMH) [55].

Since both the PFTWH and CFTWH in this work employed the same type of mass flow controllers and a room-temperature bubbler to supply water vapour, the corresponding uncertainties were consistently considered in the evaluation. Furthermore, the key signal processing filter parameters of the two systems are presented in Table 6, as they directly affect the system noise and precision.

This allowed us to estimate the accuracy, including contributions from optical path length, fitting procedure, spectral line intensity, and temperature/pressure effects, thereby supporting the reliability of the accuracy comparison results presented in Section 4.5.

In this work, we limit our discussion to accuracy and precision and comparatively analyze PFTWH with other TDLAS laboratory and airborne measurements.

To discuss the precision and accuracy requirements of PFTWH for atmospheric process, it is important to theoretically estimate the accuracy and precision of PFTWH and compare them with other TDLAS methods and our long-term measurement in Section 4.4, in order to examine whether PFTWH meets atmospheric measurement requirement. Generally, direct comparison of PFTWH with airborne measurement is challenging not just because PFTWH falls short of the standard of atmospheric process research, but it is also difficult to obtain airborne TDLAS measurement results from existing literature. We were able to find the reference of airborne TDLAS measurement from dTDLAS method - IMPMH [55], high-speed airborne instrument (HAI) [62], and stratospheric aircraft lidar for detection of humidity (SEALDH) [63–65]. Hence, we limit the object of comparison to laboratory-based TDLAS approaches that also have been applied to airborne measurement. For this analysis, we surveyed literatures from IMPMH, HAI, and SEALDH on both laboratory-based and airborne TDLAS measurement. We followed the definition of accuracy and precision of their works and applied those to our measurement results to obtain uncertainties - accuracy and precision and afterwards, we compared our values with them. The comparison was limited to results in which both accuracy and precision were reported. Works on either one or the other parameter was omitted was excluded from the comparison. For data processing in PFTWH, we used 100 Hz frequency for the measurement as presented in IMPMH work. In this study, the Allan plot was employed to determine the optimal number of averaging scans. Accordingly, the time resolution was obtained by dividing the measurement frequency (140 Hz) by the number of averaging scans, which was set to either 20 or 32.

Also, as presented on Fig. 12. and Table 7, SNR, C_{noise}, and $C_{precision}$ of both PFTWH and CFTWH were calculated by following the methods of IMPMH, HAI and SEALDH and compared with them as shown in Table 8. All calculation steps and detailed results are included in Table S2 and Table S3.

In terms of SNR, the PFTWH and CFTWH systems initially exhibited about 10–20 % of the performance observed with the IMPMH. However, since the optical paths of PFTWH (0.245 m) and CFTWH are nearly 40 times shorter than that of IMPMH (10.4 m), it is futile to make direct SNR comparisons among them. Therefore, it is more appropriate to normalize their performance in terms of precision and accuracy. From this perspective, CFTWH demonstrated the highest performance, while PFTWH and IMPMH showed equivalent results.

The observed difference can be attributed to the superior filter performance of CFTWH, which more effectively attenuates high-frequency noise components. Consequently, its 2 f signal exhibits a cleaner baseline and a more pronounced peak amplitude. In contrast, the post-processed PFTWH signal may suffer from residual noise amplification during digital filtering, leading to increased fluctuation and a reduced SNR. This difference in SNR can be also explained using the equivalent noise bandwidth (ENBW) of the filters. Since the filtering method and conditions directly affect the SNR, and the ENBW is inherently determined by the filter type and order [66].

As shown in Table 6, the Infinite Impulse Response (IIR) based low-pass filter in CFTWH exhibits a steep attenuation, effectively suppressing high-frequency noise. The smaller ENBW of this filter reduces the output noise power, resulting in higher SNR. In contrast, the FIR filter employed in PFTWH has a wider transition band and consequently a larger ENBW, which allows more noise components to pass through and leads to a relatively lower SNR. In the future, aside from processing time improvement, further enhancement of the SNR performance of the

Table 6
Filter characteristics of the CFTWH and PFTWH systems.

Parameter	CFTWH	PFTWH
Filter type	IIR low-pass filter	FIR low-pass filter 200-tap windowed-Hamming
Cut-off frequency [Hz]	300	100
Attenuation slope	12 dB/oct	gentle roll-off

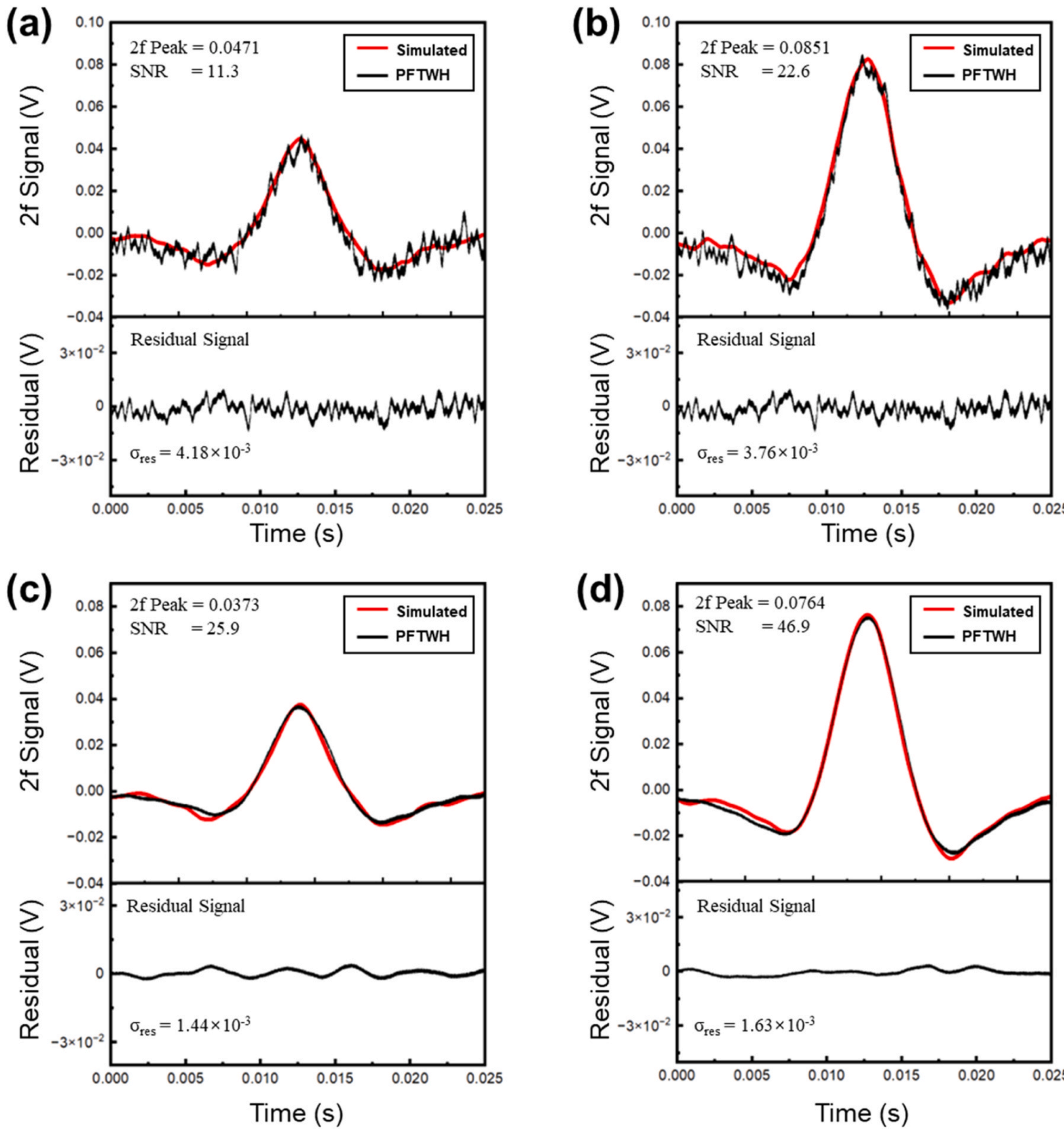

Fig. 12. Comparison of Simulated and Measured 2 f Signals and Residuals for PFTWH ((a) 2040 ppm, (b) 4920 ppm) and CFTWH ((c) 2040 ppm, (d) 4920 ppm).

Table 7
Calculation of precision using PFTWH, CFTWH and IMPMH.

	PFTWH (2040 ppm)	PFTWH (4920 ppm)	CFTWH (2040 ppm)	CFTWH (4920 ppm)	IMPMH (196 ppm)	IMPMH (12,700 ppm)
Pressure [hPa]	~ 1 atm	~ 1 atm	~ 1 atm	~ 1 atm	1010	1010
Optical path length [m]	0.245	0.245	0.245	0.245	10.4	10.4
Peak	0.0471	0.0851	0.0373	0.0764	0.149	5.57
Residual	0.00418	0.00376	0.00144	0.00163	0.000512	0.0146
SNR	11.3	22.6	25.9	46.9	291	382
$C_{absolute}$ [ppmv]	2040	4920	2040	4920	196	12,700
C_{noise} [ppmv]	181	217	78.6	105	0.674	33.3
$C_{precision}$ [ppmv m $Hz^{(-1/2)}$]	14.0	16.8	6.09	8.13	0.701	34.6
$C_{precision}$ [%]	0.688	0.342	0.299	0.165	0.358	0.273
Time resolution [Hz]	10.0	10.0	10.0	10.0	100	100
Accuracy (uncertainty)	< 4.00 %	< 4.00 %	< 4.00 %	< 4.00 %	< 4.00 %	< 4.00 %

Table 8
Summary of precision and accuracy obtained using PFTWH, CFTWH and conventional TDLAS systems (P ~ 1 atm).

	Optical path length [m]	SNR	C_{noise} [ppmv]	$C_{precision}$ [ppmv m $Hz^{(-1/2)}$]	$C_{precision}$ [%]	Accuracy [%]
PFTWH (2040 ppm)	0.245	11.3	181	14.0	0.688	< 4.00
PFTWH (4920 ppm)	0.245	22.6	217	16.8	0.342	< 4.00
CFTWH (2040 ppm)	0.245	25.9	78.6	6.09	0.299	< 4.00
CFTWH (4920 ppm)	0.245	46.9	105	8.13	0.165	< 4.00
IMPMH (196 ppm)	10.4	291	0.674	0.701	0.358	< 4.00
IMPMH (12,700 ppm)-Local	10.4	382	33.3	34.6	0.273	< 4.00
SEALDH−0 (975 hPa, 601 ppm)	1.10	140	4.30	1.79	0.300	3.90 ± 1.50 ppmv
SEALDH−0 (910 hPa, 20,000 ppm)	1.10	593	33.6	14.0	0.0700	3.90 ± 1.50 ppmv
SEALDH- II (943 hPa, 603 ppm)	1.50	344	1.75	2.31	0.160	4.30 ± 3.00 ppmv
SEALDH- II (1050 hPa, 2520 ppm)	1.50	618	4.06	2.30	0.0900	4.30 ± 3.00 ppmv
SEALDH- II (1010 hPa, 8040 ppm)	1.50	476	16.9	0.990	0.120	4.30 ± 3.00 ppmv

PFTWH system will require more sophisticated low-pass filter design methods.

It was also observed uncertainty requirements, including both accuracy and precision, of PFTWH and CFTWH are comparable to those of IMPMH, SEALDH-0 and SEALDH-II at ambient condition (~ 1 atm).

Furthermore, under variable low-pressure conditions typical of airborne measurements (HAI and SEALDH), both systems demonstrated equivalent precision and accuracy, as presented in Table S3, indicating that PFTWH and CFTWH are also suitable for airborne applications. The C_{noise} values in Tables 7 and 8 were comparable to the precision obtained from long-term measurements (Table 5) and from the Allan deviation (Fig. 9), both of which represent equivalent precision.

Overall, it is concluded that PFTWH and CFTWH show comparable performances to those of both previous laboratory-based and airborne TDLAS methods. It demonstrates that PFTWH and CFTWH meets the uncertainty - accuracy, and precision - requirements for atmospheric process research and is anticipated to be applied to airborne measurements in a broad range concentration. Also, it is expected that by extending the optical path length of current WMS configuration, PFTWH and CFTWH will meet SNR requirements for airborne measurements.

5. Conclusion

This study demonstrates the capability of the PFTWH to measure water vapour concentrations across a wide range of 70–5800 ppm, thereby extending the applicability of WMS technology from conventional low-concentration detection to broader atmospheric conditions relevant to climate science. Both the PFTWH and CFTWH employed a single near-infrared DFB laser, and FPGA-based electronics enabled fully digital demodulation of the WMS signals. By applying normalized 2 f/1 f detection, the sensitivity and robustness of the spectrometer were enhanced. In addition, the PFTWH incorporated a custom-designed DLIA with a 200-tap FIR filter, which provided linear-phase response and ensured comparable operating conditions to the CFTWH.

To validate performance, an I-WMS platform was developed, enabling operation of PFTWH, CFTWH, and dTDLAS. The linearity of the FPGA-based systems was confirmed with respect to water vapour concentration measured by dTDLAS, with coefficients of determination above 0.997 across the entire concentration range. Accuracy was maintained within 1–4 %, and at representative concentrations, the 1 f and 2 f signals from both systems showed excellent agreement, with deviations below 5 %. These results confirm that the PFTWH achieves high linearity and accuracy comparable to the CFTWH, even at the upper end of the tested concentration range.

Long-term stability was also demonstrated. The PFTWH achieved continuous measurements with 0.1-s time resolution over 1000 s, and Allan deviation analysis showed that with 30-s time-averaging the precision was measured to be 1–2 %, representing a two- to fivefold improvement. This confirms that the system is well suited for extended operation, a critical requirement for atmospheric monitoring. Furthermore, theoretical uncertainty estimates indicated that both PFTWH and CFTWH satisfy accuracy and precision thresholds required for atmospheric process research. When combined with appropriate optical path lengths, the system also has the potential to meet SNR requirements for airborne platforms.

In conclusion, the PFTWH demonstrates stable, accurate, and precise concentration measurements over a wide range, comparable to commercial systems. Importantly, the FPGA implementation offers programmability and scalability advantages, enabling flexible adaptation to different measurement conditions. Future work will investigate repeatability and robustness under varying environmental conditions, including temperature and pressure fluctuations, to strengthen applicability in diverse real-world scenarios and support broader deployment in environmental and climate science.

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CRediT authorship contribution statement

Kyunghoon Kim: Validation, Methodology. **Hongyun So:** Writing – review & editing, Resources. **Yangmo Kim:** Software. **Jeongmin Kim:** Writing – review & editing. **Seyeon Hwang:** Methodology. **Jinwoo Lee:** Writing – review & editing. **Minyoung Choi:** Writing – original draft, Validation, Conceptualization. **Gyuwon Cho:** Writing – original draft, Visualization, Data curation. **Sun Choi:** Supervision, Project administration, Funding acquisition.

Declaration of Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.snb.2025.138930.

Data availability

No data was used for the research described in the article.

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Glossary

BRAM: block random access memory
CFTWH: commercial FPGA-TDLAS-WMS hygrometer
DDS: direct digital synthesizer
DFB: distributed feedback
DLIA: digital lock-in amplifier
DSP: digital signal processor
dTDLAS: direct TDLAS
ENBW: equivalent noise bandwidth
FIR: finite impulse response
FPGA: field programmable gate array
HAI: high-speed airborne instrument
IMPMH: integrative methods and processing methods for hygrometer
IIR: infinite impulse response
I-WMS: integrated WMS
LIA: lock-in amplifier
LUT: look-up table
PFTWH: programmable FPGA-TDLAS-WMS hygrometer
PSD: phase-sensitive detection
SEALDH: stratospheric aircraft lidar for detection of humidity
SNR: signal-to-noise ratio
TDLAS: tunable diode laser absorption spectroscopy
WMS: wavelength modulation spectroscopy

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